Week 1 5/6

Ch 2 : Groups

Def : Let G be a set. A **binary operation** $*$ on G is a function that assigns to **each ordered pair of elements in G** an element **in G**.
(pg31)

Input $\in G$, output $\in G$

We write : $* : G \times G \to G$

$(a, b) \to a * b$

where $G \times G : \{(a, b) \mid a, b \in G\}$ is the Cartesian Product.

eg ① : $G = \mathbb{Z}$, then $* = +$ is a binary op

i.e $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$

$(a, b) \to a + b$

e.g. ② : $G = \mathbb{R}$, Then $* = \cdot$ ← usual multi is a bin op on $\mathbb{R}$

e.g. ③ : $Sym(\not\Delta)$, Then $* = \circ$ is a bin op here

i.e. $\forall f, g \in Sym(\not\Delta)$, $f \circ g \in Sym(\not\Delta)$

e.g. ④ : $G = M_n(\mathbb{R})$ the $n \times n$ real entry matrices.

then set $*$ be matrix multiplication or matrix addition are bin op on G.

Def : A **group** is a pair $(G, *)$ of a **non-empty set** G and a **binary operation $*$** on G satisfying.
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Abstract Def of Groups

1. Associativity : $\forall a, b, c \in G$, $(a * b) * c \Rightarrow a * (b * c)$
2. Identity axiom : $\exists e \in G$, $\forall g \in G$ s.t $e * g = g = g * e$

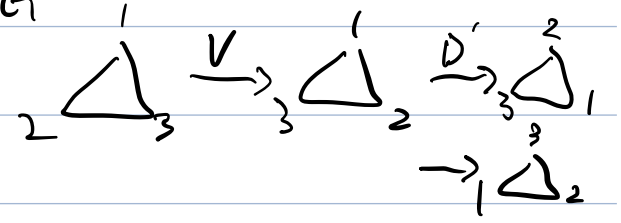
change order then it will gives another math def.

3. Inverse axiom : $\forall g \in G$, $\exists g^{-1} \in G$ s.t $g * g^{-1} = e = g^{-1} * g$

5 def : non-empty + binary operation + 3 Abstract Def.

Conventions : We will write / denote our group as G

We will write $a * b$ as ab



Ex : $(Sym(\triangle), \circ)$ is a group.

Pf: Recall : $Sym(\triangle) = \{R_0, R_{120}, R_{240}, V, D', D\}$ (D: diagonal reflection)

↑ 0-degree rotation, rotation always counter-clockwise.

- Note : $R_0 \in Sym(\triangle)$ $\therefore Sym(\triangle) \neq \emptyset$, non empty ✓
- We know $\forall f, g \in Sym(\triangle)$, $f \circ g \in Sym(\triangle)$ (checked another def → (Bin op))

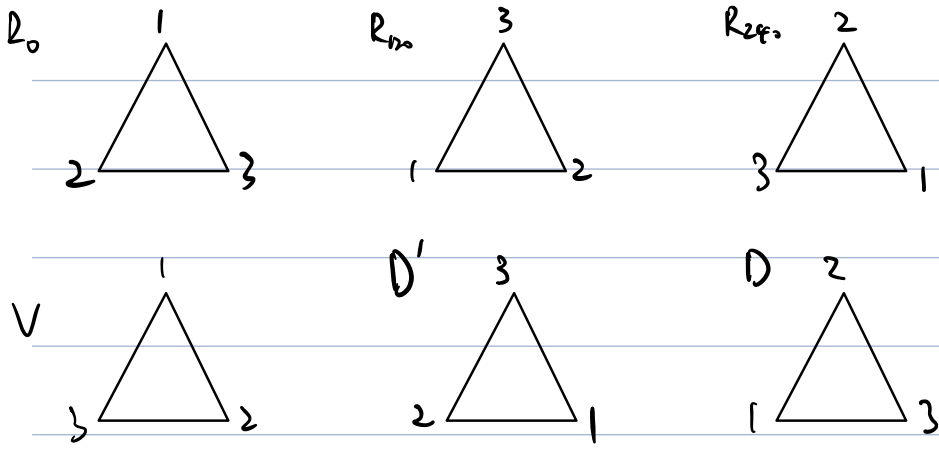
  by our prior exercise, Namely when we filled out the Cayley Table for $(Sym(\triangle), \circ)$
- Choose $e = R_0 \in Sym(\triangle)$ (check Cayley Table first row, first column)

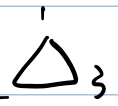
  Observed from the first row and first column of our Cayley Table, we get $\forall g \in Sym(\triangle)$, $R_0 \circ g = g = g \circ R_0$

  $\therefore Sym(\triangle)$ has a 2-sided id (Inverse Axiom)
- From the prerequisite materials, composition is associative, so $(Sym(\triangle), \circ)$ (When defined) inherits this property.

Cayley Table :

$\circ$	R_0	R_{120}	R_{240}	V	D'	D
R_0	R_0	R_{120}	R_{240}	V	D'	D
R_{120}	R_{120}	R_{240}	R_0	D	V	D'
R_{240}	R_{240}	R_0	R_{120}	D'	D	V
V	V	D'	D	R_0	R_{120}	R_{240}
D'	D'	D	V	R_{240}	R_0	R_{120}
D	D	V	D'	R_{120}	R_{240}	R_0



$R_0\ R_0$  . $R_0 V = V$, $R_{120} V = \begin{matrix} 3 \\ 2 \end{matrix} \begin{matrix} \times \\ \times \end{matrix} \begin{matrix} 1 \\ 1 \end{matrix} = D'$ $\begin{matrix} 1 \\ 3 \\ 2 \end{matrix} \begin{matrix} 2 \\ 1 \\ 3 \end{matrix} = D$

$R_{120} \circ V$ D_0 V first and then R_{120}

$(f \circ g)(x) = f(g(x))$ ⇐ $\circ$ means composite

- Each element in $Sym(\diamondsuit)$ has a 2-sided inverse ✓

  Namely: R, V, D, D' are their own inverse.

  - R_{180} has R_{360} as it's 2-sided inverse and vice versa.

□

Ex: Is $(\mathbb{N}, -)$ a group?

↖ subtraction

Sol: No!, subtraction is not a bin op on $\mathbb{N}$ (also fail other rules)

WTS: $\forall a, b \in \mathbb{N}$, $a - b \in \mathbb{N}$ is false.

=((WTS: $\exists\, a, b \in \mathbb{N}$, $a - b \in \mathbb{N}$ is true.

Choose $a = 1$, $b = 2$, observe $a, b \in \mathbb{N}$

But $a - b = 1 - 2 = -1 \notin \mathbb{N}$

□

Basic Properties:

Thm: If G is a group, then it's identity is unique.
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Pf: Suppose G is a group.

Assume e, e' are identies in G

i.e. $e, e' \in G$, $\forall g \in G$, st. $eg = g = ge$ ① and $e'g = g = ge'$ ②

WTS: $e = e'$

Notes: ① $\Rightarrow$ $ee' = e'$ ($g = e'$)

they are equal

② $\Rightarrow$ $e = ee'$ ($g = e$)

$\Rightarrow e = e'$ as we wanted □

Thm: Let G be a group. Then each $g \in G$ has a unique (2-sided) inverse.

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Pf: exercise $\quad g * g^{-1} = e = g^{-1} * g$

Set $g * a = e = a * g$, $\quad g * b = e = b * g$

Set $a = ae \qquad a = a(gb) = (ag)b \Rightarrow ae = eb \Rightarrow a = b$

by associativity

Thm: Let G be a group. Then

pg 38.

① $\forall a, b, c \in G$, if $ab = ac$ then $b = c$ (left cancellation law)

② $\forall a, b, c \in G$, if $ba = ca$ then $b = c$ (right cancellation law)

Pf: Let G be a group — Detail Justification ☆

Let $a, b, c \in G$ be arbitrary

Suppose $ba = ca \Rightarrow$ Cancel out a then we need a^{-1}

WTS: $b = c$

Note since G is a group, every element in G has an inverse.

In particular, $a' \in G$ is an inverse of $a \in G$

Now $ba = ca \Rightarrow (ba)a' = (ca)a'$ by associative

$\Rightarrow b(aa') = c(aa')$ by 2-sided

$be = ce$ by Identity axiom of G

$b = c$

∎

Week 2 TUT

Binary Operations :

Ex 1 : Is multiplication mod 9 a bin op on $\{1, 4, 5, 7\}$?

Sol : No , $4 \cdot 5 = 20 \equiv 2 \pmod 9$ not in $\{1, 4, 5, 7\}$

Ex 2 : Is $\cdot$ a bin op on $H = \{n^2 \mid n \in \mathbb{Z}\}$?

Sol : $n \in \mathbb{Z}$ $n^2 \in \mathbb{Z}$ $n^2 \cdot m^2 \in \mathbb{Z}$ It is bin op.

$m = a^2$, $n = b^2$ $m \cdot n = a^2 \cdot b^2 = (a \cdot b)^2 \in H$

Ex 3 : Group or not ? $G = \{f_{a,b}(x) = ax + b \mid a \in \mathbb{R}^\times = \mathbb{R} - \{0\}, b \in \mathbb{R}\}$ under function composition

Group : ① Non - Empty , $G = f_{a,0}(x) = 0$ ✓

② binary op ,

③ $ax+b + cx+d = cx+d + ax+b$ ✓

④ ✓

⑤ $g * g^{-1} = e = g^{-1} * g$

Yes it is group.

- non - empty $f_{1,0}(x) = x \in G$
- closure let $f_{a,b}(x) = ax+b$, $f_{c,d}(x) = cx+d$ where $a, c \in \mathbb{R}^\times$, $b, d \in \mathbb{R}$

  $(f_{a,b} \circ f_{c,d})(x) = f_{a,b}(f_{c,d}(x)) = a(cx+d) + b = (ac)x + ad+b$

  $ac \in \mathbb{R}^\times$, b/c $a \neq 0$ and $c \neq 0$
- associativity //
- identity claim $f_{1,0}(x) = x$ is the id , verify $f_{1,0} \circ f = f = f \circ f_{1,0}$ ∀f∈G
- inverse. let $f_{a,b} = ax + b$ Suppose $f_1^{-1}(x) = cx + d$

  We know $f_1^{-1} = x \Rightarrow (ac)x + (ad+b) = x$

$$\Rightarrow \begin{cases} ac = 1 \\ ad+b = 0 \end{cases} \Rightarrow \begin{cases} c = \frac{1}{a} \\ d = -\frac{b}{a} \end{cases} \Rightarrow f_1^{-1}(x) = \frac{1}{a} - \frac{b}{a}$$

exercise : Show f_1^{-1} exists b/c $a \neq 0$

Verify f_1^{-1} is the inverse of f_1, i.e. $f_1^{-1} \circ f_1 = x = f_1 \circ f_1^{-1}$

Ex4: Let $G = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{Z}_4, \ ad - bc \neq 0 \pmod 4 \right\}$

Is G a group under matrix multiplication?

- Non-empty 4×0, 4×1, 4×2, 4×3 ✓
- Closure : $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & j \end{bmatrix} = \begin{bmatrix} ae+bg & af+bj \\ ce+dg & cf+dj \end{bmatrix} \in \mathbb{Z}_4$ ✓
- ?
- $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} = [0]$ ✓
- $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = [I]$ ✓

Not a group. $\mathbb{Z}_4$ has zero divisors.

$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ $\det(A) = 2 \mod 4 \neq 0$

$\det(A \cdot A) = 4 \mod 4 = 0 \Rightarrow$ Not close.

∴ This set is not closed under multiplication.

$GL(2, \mathbb{Z}_p)$ forms a gp under matrix mult.

Ex5: Why is $|D_n| = 2n$?

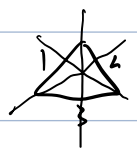
$D_3 = \{$ Symmetries of the equivlateral triangle $\} = Sym(\Delta) \Rightarrow |D_3| = 6$

$D_4 = 8$ Vertex 1 has 4 destinations

After fixing vertex 1, 2 ways to orient the adjacent vertices.

(keeping the clockwise order, or reversing it via reflection)

Total Symmetry $4 \times 2 = 8$





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Def: Let G be a group.

- The order of G is the size of G. This is written as $|G|$.
- The order of $g \in G$ is the smallest positive number of times n.

  s.t. $g^n = \underbrace{g \cdot g \cdot g \cdots}_{n \text{ times}} = e$

If no such n exists then we say g has **infinite orders**. Denoted $|g|$

- We say G is **abelian** if $\forall a, b \in G$, $ab = ba$

Ex: Consider $\text{Sym}(\hexagram) = \{R_0, R_{120}, R_{240}, V, D', D\}$

What $|\text{Sym}(\hexagram)|$, $|R_{120}|$? Is $\text{Sym}(\hexagram)$ abelian or not?

Soln: Note $|\text{Sym}(\hexagram)| = 6$

Observe $R_{120} \neq R_0 \quad \therefore |R_{120}| \neq 1$

$R_{120}^2 = R_{120} R_{120} = R_{240} \neq R_0$

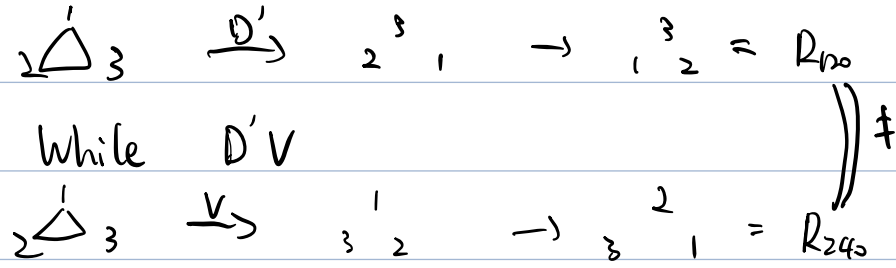
$\therefore |R_{120}| \neq 2$

$R_{120}^3 = R_{120} R_{120} R_{120} = R_0 \Rightarrow |R_{120}| = 3$

Claim: $\text{Sym}(\hexagram)$ is not abelian

Choose $V, D' \in \text{Sym}(\hexagram)$

Notice VD'



While $D'V$

$\neq$

□

Examples of Groups:

① Some sets of symmetries w. composition.

Prop: The plane symmetries of any 2-D figure w. composition from a gp

e.g. $\text{Sym}(\hexagram)$, $\{R_0, R_{90}, R_{180}, R_{270}, V, H, D', D\}$

Note: The group of plan symmetry of regular n-gon is called the Dihedral Group. We will denote D_n.

Prop: The rotational symmetries of any 3-D figure w-composition is a gp.

2. Some sets of #'s w. addition

e.g. $(\mathbb{R}, +)$ Pf: $\mathbb{R} \neq \emptyset$ b/c $0 \in \mathbb{R}$

- we know "+" is a bin op, and associativity hold by axioms of $\mathbb{R}$.
- WTS. $\exists\, e \in G$, $\forall g \in G$ s.t. $ge = g = eg$

  Choose $e = 0 \in \mathbb{R}$, let $g \in \mathbb{R}$ be arb

  Then $g + 0 = g$ and $0 + g = g$
- WTS $\forall g \in G$, $\exists\, g^{-1} \in G$ st. $gg^{-1} = e = g^{-1}g$

  Let $g \in \mathbb{R}$ be arb. choose $-g \in \mathbb{R}$ by def of $\mathbb{R}$

  Then $g + (-g) = 0 = -g + g$

Exercise: $(\mathbb{Z}, +)$ $(\mathbb{Q}, +)$, $(\mathbb{C}, +)$

③ Some sets of #'s w. mult.

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We seen

① Some sym. w - composition

② Some sets of #'s w - "+"

③ Some sets of #'s w - "·"

④ Some sets of matrices w. matrix mult.

Define $GL(2,\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid \begin{matrix} a,b,c,d \in \mathbb{R} \\ ad-bc \neq 0 \end{matrix} \right\}$ is called the general linear group.

Prop: $GL(2,\mathbb{R})$ is a group of w. matrix mult. Pf: exercise

⑤ Permutations

Def: Let $X \neq \phi$. A permutation on X is a bijection $f: X \to X$

Pg19

e.g.: Let $X \neq \phi$. The function maps himself $\Rightarrow \alpha(x) = x$, $\forall x \in X$, s.t $\alpha: X \to X$ is a permutation.

proposition: The set $Perm(X) = \{ f: X \to X \mid f \text{ is a bijection} \}$ w. composition is a group.

Pf: WTS: ① $Perm(X) = \phi$

② Composition be bin op

③ Associativity

④ Two sided Identity

⑤ Two sided inverse.

①: Note $f(x) = x$ is bijection on X.

$\therefore Perm(X) \neq \phi$

②: WTS $\forall f, g \in Perm(X)$, $f \circ g \in Perm(X)$

Let $f, g \in Perm(X)$ be arb, WTS $f \circ g : X \to X$ is a bijection

$\Rightarrow f \circ g$ is 1-1 (one to one)

Way 1: recall a fcn h is injective, iff $\forall a, b \in dom(h)$, if $h(a) = h(b)$ then $a = b$

$f \circ g : X \xrightarrow{g} X \xrightarrow{f} X$

Let $x_1, x_2 \in X$ be arb.

Suppose $f(g(x_1)) = f(g(x_2))$

How to "kill" f and g $\Rightarrow x_1 = x_2$? Hint: use bijective and Injective.

Soln: $g(x_1) = g(x_2)$ b/c f is 1-1 as $f \in Perm(X)$

$x_1 = x_2$ b/c g is 1-1 as $g \in Perm(X)$

Onto: recall h is onto, iff $\forall x \in codom(h)$, $\exists z \in dom(h)$ s.t $h(z) = x$

Let $x \in X$ be arb

We know f is onto b/c $f \in Perm(X)$ so $\exists y \in X$ s.t $f(y) = x$

We also know g is surjective (g is onto as $g \in Perm(X)$)

$\therefore \exists z \in X$ s.t $g(z) = y$

$X \xrightarrow{g} X \xrightarrow{f} X$

$z \longrightarrow y \longmapsto x$

Thus we have $\exists z \in X$ and $f(g(z)) = f(y) = x$

$\therefore f \circ g : X \to X$ is bijection

i.e. $f \circ g \in Perm(X)$

Exercise: Prove ③ ④ ⑤

Ex: Let $X = \{1, 2, 3\}$

List all members of $Perm(X)$

Soln: f_1 $1 \to 1$

$2 \to 2$

$3 \to 3$

f_2 $1 \to 2$

$2 \to 3$

$3 \to 1$

f_3 $1 \to 3$

$2 \to 1$

$3 \to 2$

f_4: $1 \to 1$

$2 \to 3$

$3 \to 2$

f_5: $1 \to 3$

$2 \to 2$

$3 \to 1$

f_6: $1 \to 2$

$2 \to 1$

$3 \to 3$

These are bijections by construction and this is all of them.

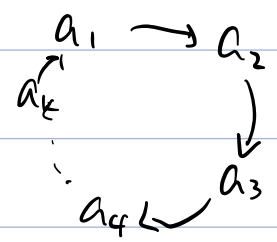
i.e. $\text{Perm}(X) = \{f_1, f_2, f_3, f_4, f_5, f_6\}$

Def: Let $n \in \mathbb{Z}^+$. Let $X = \{1, 2, 3 \cdots n\}$

pg82 Suppose $a_1, a_2 \cdots a_k$ are distinct elements in X, $k \leq n$. A cycle or k-cycle or a cycle of length k is a function, $a_1 \to a_2$, $a_2 \to a_3 \cdots a_{k-1} \to a_k$ $a_k \to a_1$. All other elements are fixed.

Denoted $(a_1, a_2, a_3 \cdots a_k)$

Geometrically:



f_1: (1), (2), (3) := (1)

Mapping to himself.

f_2: (1, 2, 3)

f_3: (3, 2, 1)

f_4: (2, 3)

f_5: (1, 3)

f_6: (1, 2)

Notice: (1, 2, 3) = (2, 3, 1) = (3, 1, 2)

Ex: Consider: $f_2 = (1,2,3)$, $f_3 = (3,2,1)$ from prior example.

Compute $f_2 \circ f_3 = (1,2,3)(3,2,1)$

Soln: $1 \to 3 \Rightarrow f_3(1) = 3$

$1 \xrightarrow{f_3} 3 \xrightarrow{f_2} 1 := f_2(f_3(1)) = f_2(3) = 1$

$2 \to 1 \to 2$

$3 \to 2 \to 3$

$f_2 \circ f_3 = (1)(2)(3)$

$= (1)$

Ex: Compute $(2\ 3)(1\ 2\ 3)$

$= (1\ 3)(2)$

$= (1\ 3)$

Exercise: Write out Cayley Table in Cycle Notation

Compute the Cayley Table for Perm(X) in prior Example in Cycle Notation

Def: Let $n \in \mathbb{Z}^+$. Let $X = \{1, 2, \cdots n\}$

pg81 The group of permutations on X is called the <u>symmetric group</u> of degree n Denoted S_n.

Modular Arithmetic

For any $a \in \mathbb{Z}$, $n \in \mathbb{Z}^{>0}$, by the division algorithm. $\exists q, r \in \mathbb{Z}$ s.t. $a = qn + r$, where $0 \leq r < n$

Def: let $a \in \mathbb{Z}$, $n \in \mathbb{Z}^{>0}$. We say a is congruent to r modulo n denoted $r \equiv a \bmod n$, if $n \mid r-a$

Def: The set $\mathbb{Z}_n = \{r \mid a \equiv r \bmod n\} = \{0, 1, 2 \cdots n-1\}$ is called the integers module n.

e.g. $8 \bmod 4 \equiv 0$

$11 \bmod 5 \equiv 1$

$-1 \bmod 4 \equiv 3$

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Recall, let $n \in \mathbb{Z}^+$. We defined $\mathbb{Z}_n = \{ r \mid r \equiv a \bmod n,\ a \in \mathbb{Z} \}$

integers modulo n = $\{0, 1, 2, 3, \dots, n-1\}$

Def $U(n) = \{ a \in \mathbb{Z}_n \mid \gcd(a, n) = 1 \}$ is called the units modulo n.

Note: we can define let $a, b \in \mathbb{Z}_n$

- addition mod n $\quad a +_n b : (a + b) \bmod n$ ← in usual
  ↖ show closure.
- mult mod n : $a \cdot_n b : (ab) \bmod n$

Ex: Consider $\mathbb{Z}_4 = \{0, 1, 2, 3\}$

Examine

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$1, 3 \in \mathbb{Z}_4$

$1 +_4 3 = 4 \bmod 4$

$= 0$

↑ Group

$\cdot_4$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

↑ Not a group.

Remark: fix $n \in \mathbb{Z}^+$

- $(\mathbb{Z}_4, +_4)$ is a group. — Exercise: proof ✡
- $(\mathbb{Z}_4, \cdot_4)$ is not a group — Exercise: proof ✡
- $(\{1, 3\}, \cdot_4)$ is a group — Exercise

  ↑ $U = U(4)$

End for Quiz 1, can finish all A and B

Ch3: Sub groups.

Def: pg67 Let G be a group. A sub group H of G is a non-empty subset of G that is itself a group under the same binary operation as G.

Notation $H \subseteq G$

fast eg: ① let G be a group.

- $G \subseteq G$
- $H = \{e\} \subset G$ ← the trivial subgroup of G

Remark a sub group H of H s.t $H \neq G$, we call a proper subgroup. Denoted $H \subset G$

② $H = \{R_0, R_{120}, R_{240}\} \subset D_3$

③ $(H, +) = (\mathbb{Z}^\times, \cdot)$ is Not a sub group of $(\mathbb{R}^*, \cdot)$ [Why?]

④ $(H, \cdot) = (\mathbb{Z}, \cdot)$ is Not as $(\mathbb{R}, +)$

(Not same bin op.)

Even change $(\mathbb{Z}, \cdot)$ to $(\mathbb{Q}^\times, \cdot)$, $(\mathbb{Q}^\times, \cdot)$ is a group.

Thm: One-step subgroup test. pg68

Let G be a group. Suppose $H \subseteq G$ (subset), s.t. $H \neq \emptyset$ (non empty)

If $\forall a, b \in H$, $ab^{-1} \in H$

THEN $H \leq G$

Pf: Given $(G, *)$ is a group. ①

Suppose $H \subseteq G$ ② and $H \neq \emptyset$ ③, * Assume $\forall a, b \in H$, $ab^{-1} \in H$ ··· ④

WTS $H \leq G$ (Proceed by def)

Observe ②, ③ give us two requirements in def of subgroup.

- Associativity ② ⇒ $(H, *) \subseteq (G, *)$

  We know H is assoc under inherited $*$ op as G is assoc under $*$.

- Id axiom on H ; ③ ⇒ $\exists\, x \in H$

  Consider $a = b = x$

  ⇒ $x\, x^{-1} \in H$ by ④

  (x^{-1} in G)

  ∴ $e \in H$ is a 2-sided id

  b/c e is unique.

- Inverse in H, WTS $\forall\, h_1 \in H,\ h_1^{-1} \in H$ (2-sides)

  Let $h_1 \in H$ be arb. Take $a = e$, $b = h_1$

  By ④, $ab^{-1} = e h_1^{-1} \in H$

  ($= h_1^{-1}$)

  ∴ $h_1^{-1} \in H$ is 2-sided [WHY?]

  b/c $h_1^{-1} \in G$ and G has unique inverse elements.

- Closure in H ; WTS $\forall\, h_1, h_2 \in H,\ h_1 * h_2 \in H$.

  Let $h_1, h_2 \in H$ be arb

  From ④ ⇒ $ab^{-1} = h_1 (h_2^{-1})^{-1} \in H$

  (take $a = h_1$, $b = h_2^{-1}$) $= h_1 h_2$ b/c $(a^{-1})^{-1} = a$

  $\in H$ $\in H$

Ex: Consider $GL(2, \mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid \begin{matrix} a, b, c, d \in \mathbb{R} \\ \det(A) \neq 0 \end{matrix} \right\}$

Prove $H = \{ A \in GL(2, \mathbb{R}) \mid \det(A) = 1 \} \subseteq GL(2, \mathbb{R})$

Pf: We want to apply one-step test.

- $H \subseteq G$ holds by def of H as $A \in GL(2, \mathbb{R})$
- $H \neq \emptyset$ b/c $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in H$

- Let $A, B \in H$ be arb.

  Consider AB^{-1}

  From linear algebra, matrix mult $\Rightarrow AB^{-1}$ is a 2×2 real entry matrix.

  Moreover, $\det(AB^{-1}) = \det(A)\det(B^{-1})$

  $= \det(A)\frac{1}{\det(B)}$

  $= 1$ b/c $A, B \in H$

$\therefore AB^{-1} \in H$, by def of H

$\therefore H \subseteq GL(2,\mathbb{R})$ ☑

Thm: two-step Subgroup Test.
pg50

Let G be a group, suppose $H \subseteq G$ and $H \neq \emptyset$

If: (i) $\forall a,b \in H$, $ab \in H$

(ii) $\forall a \in H$, $a^{-1} \in H$

THEN $H \subseteq G$

Pf: exercise (1) by def

(2) by one-step test

Def
pg53: Let G be a group.

The centre of G

$Z(G) := \{ g \in G \mid gx = xg, \forall x \in G \}$

i.e. the set of elements in G that commute w. everything in G.

Thm: Let G be a group

The $Z(G) \subseteq G$

TUT 2 Week 4 May 25

1. Given elements in S_5, compute $(1,2,3)(3,4,5)$ and find the inverse of $(1,2,3,4,5)$

Soln: $(1,2,3)(3,4,5) = (1\ 2\ 3\ 4\ 5)$

$(1,2,3,4,5)^{-1} = (5,4,3,2,1) = (1,5,4,3,2)$

2. Let G be a group and H and K be subgps of G. Prove that $H \cap K \leq G$

Soln: We use 2-step test.

① Inclusion: since $H \leq G$ and $K \leq G$, $H \cap K \leq G$

② Non-empty: Since $H \leq G$ and $K \leq G$, the id $e \in H \cap K$ (uniqueness of id used here)

③ Closure: Take $a, b \in H \cap K$. By def, $a, b \in H$ and $a, b \in K$ As subgps closed under operator $ab \in H$ and $ab \in K \Rightarrow ab \in H \cap K$

④ Inverses: Take $a \in H \cap K$. By def, $a \in H$ and $a \in K$. As subgps contain inverses $a^{-1} \in H$ and $a^{-1} \in K \Rightarrow a^{-1} \in H \cap K$

$\therefore$ By 2-step test, $H \cap K \leq G$ ∎

Recall $Z(G) = \{x \in G \mid xg = gx, \forall g \in G\}$

3. Find $Z(GL(2,\mathbb{R}))$

Soln: $\subseteq$: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in Z(GL(2,\mathbb{R}))$

Pick simple matrices.

Test Matrix 1. $E_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ Set $AE_1 = E_1A$

$\begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix} \Rightarrow c = 0,\ a = d$

A should look like $\begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$

Test Matrix 2. $E_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ Set $AE_2 = E_2A$

$\Rightarrow \begin{bmatrix} a+b & b \\ a & a \end{bmatrix} = \begin{bmatrix} a & b \\ a & a+b \end{bmatrix} \Rightarrow b = 0$

So $A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$, where $a \in \mathbb{R}^\times$ Thus $Z(GL(2,\mathbb{R})) \subseteq \{aI \mid a \in \mathbb{R}^\times\}$

$\supseteq$: Let $A = aI$, where $a \in \mathbb{R}^\times$, Let $B \in GL(2,\mathbb{R})$

$AB = (aI)B = a(IB) = aB$ and $BA = B(aI) = (Ba)I = aBI = aB$

So $AB = BA \quad \forall B \in GL(2,\mathbb{R})$

$\therefore Z(GL(2,\mathbb{R})) = \{aI \mid a \in \mathbb{R}^\times\}$

4. Let G be a gp. Suppose $a \in G$ is the only element of order 2.
Prove that $a \in Z(G)$

Let $g \in G$ WTS $ga = ag \iff gag^{-1} = a$

$(gag^{-1})^2 = (gag^{-1})(gag^{-1}) = ga(g^{-1}g)ag^{-1} = ga^2g^{-1} = geg^{-1} = e$

$\Rightarrow |gag^{-1}| \leq 2$

Could $gag^{-1} = e$? If so, $a = g^{-1}g = e$, so impossible as $|a| = 2$

$\Rightarrow |gag^{-1}| = 2 \Rightarrow gag^{-1} = a \Rightarrow ga = ag \Rightarrow a \in Z(G)$ ☑

5. Prove G is abelian if every non-identity element has order 2.

Pf: If $x^2 = e$, $x = x^{-1} \quad \forall x \in G$

Let $a, b \in G$ By closure, $a, b \in G$

$ab = (ab)^{-1} = b^{-1}a^{-1}$

↓ by scalar shoes prop

Week 4 Monday Lec

Let $(G, *)$ be a gp

Let $(H, *) \subseteq (G, *)$

How can we prove $H \subseteq G$

- by def
- 1-step test
- 2-step test

Recall $Z(G) = \{ g \in G \mid gx = xg, \forall x \in G \}$

Ex pg 53 Prove $Z(G) \subseteq G$

Pf: (2-step test)

- $Z(G) \subseteq G$ by def of $Z(G)$
- $Z(G) \neq \emptyset$ b/c $e \in G$ and $ex = xe$ $\forall x \in G$ by def of "e" (identity axiom)
- let $g_1, g_2 \in Z(G)$. WTS $g_1 g_2 \in Z(G)$

(i) Note $g_1 \cdot g_2 \in G$ b/c $g_1 \cdot g_2 \in Z(G)$ by def

$\therefore g_1 \cdot g_2 \in G$ b/c G is gp (closure)

(ii) WTS: $\forall x \in G$, $g_1 x = x g_1$, Let $x = G$

Consider $(g_1 \cdot g_2) x = g_1 (g_2 x)$ by assoc of G

$= g_1 (x g_2)$ b/c $g_2 \in Z(G)$

$= (g_1 x) g_2$ by assoc of G

$= x g_1 g_2$ b/c $g_1 \in Z(G)$

$\therefore g_1 g_2 \in Z(G)$

- Let $g \in Z(G)$ WTS $g^{-1} \in Z(G)$

(i) Note $g \in G$, b/c $g \in Z(G)$ by def

We know $g^{-1} \in G$ b/c G a gp ("Inverse")

(ii) Let $x \in G$, WTS: $g^{-1} x = x g^{-1}$

But $gx = xg$ b/c $g \in Z(G)$

=> Apply inverse to both side

$g^{-1}gx = g^{-1}xg$

$x = g^{-1}xg$ by assoc.

$xg^{-1} = g^{-1}xgg^{-1}$

$exg^{-1} = g^{-1}xe$

$xg^{-1} = g^{-1}x$

$\therefore g^{-1} \in Z(G)$, $Z(G) \leq G$ ✓

Exercise: Let G be group.

(i) prove if G is abelian then $G = Z(G)$

(ii) TIF $Z(G) = G$ iff G is abelian

Ex: Let G be a gp let $a \in G$. the set $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$

Prove $\langle a \rangle \leq G$

$n > 0$ $a^n = a \cdot a \cdots a$

$n < 0$ $a^n = (a^{-1})^{|n|}$

$n = 0$ $a^n = e$

Examples: $G = \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

- Consider $2 \in \mathbb{Z}_6$

$\langle 2 \rangle = \{(n^2) \bmod 6 \mid n \in \mathbb{Z}\}$

$= \{0, 2, 4\}$

- Consider $5 \in \mathbb{Z}_6$

$\langle 5 \rangle = \{(n5) \bmod 6 \mid n \in \mathbb{Z}\}$

$= \{0, 5, 4, 3, 2, 1\}$

Def. Given a gp G. Let $a \in G$ Then $\langle a \rangle$ is called a cyclic subgp of G (generate by a)

If $\exists\ a \in G$ s.t. $\langle a \rangle = G$, then we call G is cyclic.

Pf: of $\langle a \rangle \subseteq G$

- $\langle a \rangle \neq \emptyset$ b/c $a \in \langle a \rangle$
- $\langle a \rangle \subseteq G$ b/c $a^n \in G$ by closure as $a \in G$.

Let $a^n, b^m \in \langle a \rangle$ so $m, n \in \mathbb{Z}$

Consider $(a^n)(a^m)^{-1} = a^{n-m} \in \langle a \rangle$

TUT Week 5

1. Prove that every cyclic gp is abelian

Pf: Let $G = \langle a \rangle$, Take $x, y \in G$, by def of cyclic gp.

$x = a^n$ and $y = a^m$ for some $n, m \in \mathbb{Z}$.

Compute $x \cdot y = a^n a^m = a^{n+m}$ by assoc

$= a^{m+n}$ by integer addition

$= a^{mn}$ by assoc

$= yx$

□

2. Find $|\alpha|$ where $\alpha = (1, 3, 4, 5, 6, 2) \in S_{48}$

$|\alpha| = 6$

- $\alpha^6 = e$ ($1 \to 3 \to 4 \to 5 \to 6 \to 2 \to 1$ cyclic back)
- For $k < 6$, $\alpha^k(1) \neq 1 \Rightarrow \alpha^k \neq e$

Note that k-cycle has order k

Recall prop(i), $a^k = e \Rightarrow |a| \mid k$

3. Suppose $a^{28} = e$ and $a^{21} = e$ What can we say about $|a|$?

Let $k = |a|$. By prop(i), $k \mid 28$ and $k \mid 21 \Rightarrow k \mid \gcd(28, 21) = 7 \Rightarrow |a| \in \{1, 7\}$

4. Let G be a group, Let $a, b, g \in G$

(a) Prove that $|gag^{-1}| = |a|$

(b) Prove that $|ab| = |ba|$

Hint prove $|x| = |y|$, show that $|x| \mid |y|$ and $|y| \mid |x|$

(a) Notice $(gag^{-1})^k = ga^kg^{-1}$

Let $n = |a|$ and $m = |gag^{-1}|$

$m \mid n$: $\langle gag^{-1} \rangle^n = ga^ng^{-1} = geg^{-1} = e \Rightarrow m \mid n$, by prop 1(i)

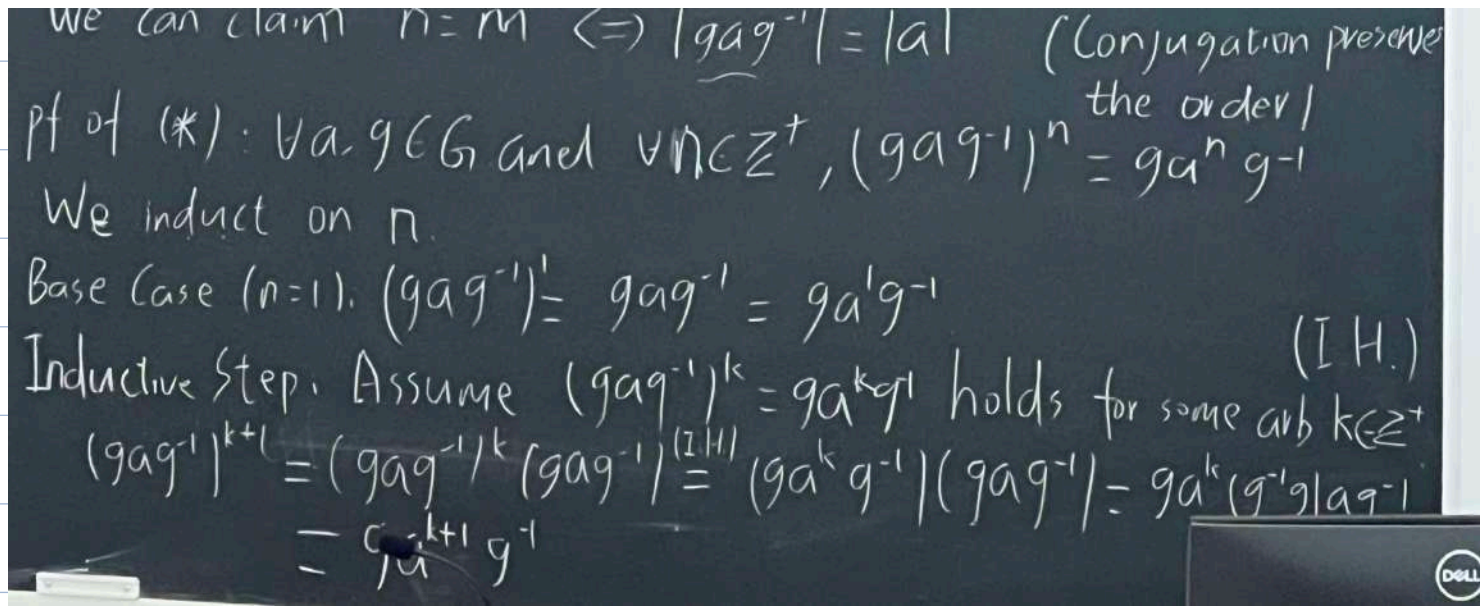
$n \mid m$: We know that $(gag^{-1})^m = e \Rightarrow ga^mg^{-1} \Rightarrow g^{-1}ga^mg^{-1}g = g^{-1}eg$

$\Rightarrow a^m = e$ Since $|a| = n$, by prop. 1(i). $n \mid m$

Since $n|m$ and $m|n$ and they are both positive integers.

We can claim $n=m \iff |gag^{-1}| = |a|$ (conjugation preserve the order)

Pf of (*)



By PMI $(gag^{-1})^n = ga^ng^{-1}$ $\forall n \in \mathbb{Z}^+$

(b) $ab = a(ba)a^{-1}$

5. If $|G|$ is even, G has element of order 2.

Pf: Every element $x \in G$ has a unique inverse. If we pair them together $\{x, x^{-1}\}$ if $x \neq x^{-1}$ then $\{x, x^{-1}\}$ has size of 2.

We partition the entire group G using $\{x, x^{-1}\}$

Since $e = e^{-1}$, we take e out.

Since $|G|$ is even, $|G|-1$ is odd. If we try to pair up $\{x, x^{-1}\}$, there must be an odd number of non-identity elements that cannot be paired with their distinct inverse. That means, $\exists\, a \in G$, s.t. $a \neq e$ and $a = a^{-1} \Rightarrow a^2 = e$ and $a^2 \neq e$

$\Rightarrow |a| = 2$ ☑

Lec Week 5.

Recall: **Prop 1**: Let G be a group, Let $a \in G$ st. $|a| = n$ some $n \in \mathbb{Z}^+$

Then:

(i) If $a^k = e$, for any $k \in \mathbb{Z}$ then $n | k$

(ii) $|a| = |\langle a \rangle|$

Pf: (ii) Suppose set-up

Recall $\langle a \rangle = \{a^m \mid m \in \mathbb{Z}\}$

By div-alg, $\exists\, q, r \in \mathbb{Z}$ s.t $m = qn + r$ where $0 \leq r < n$

$\therefore a^m = a^{qn+r}$

$= (a^n)^q a^r$ by assoc

$= e^q a^r$ blc $a^n = e$

$= a^r$ blc e is the id

Then $\langle a \rangle = \{e, a^1, a^2, a^3 \cdots a^{n-1}\}$ are these elements distinct?

If $a^i = a^j$, for some $0 \leq i \leq j < n$

$\Rightarrow e = (a^i)^{-1} a^j = a^{j-i}$ by gp axioms (inverse of assoc)

$\Rightarrow$ Why $j - i = 0$? blc $|a| = n$ $0 \leq j - i < n - i < n$

$\Rightarrow j = i$ $\therefore$ all a^r distinct

Prove a^r distinct is $j - i = 0$ has to be true or $j - i = 0$ is an assumption?

Ex: Let G be a gp, Let $a \in G$.

Prove If $a^{17} = e$ and $a^{25} = e$ then $a = e$

Pf: Suppose set-up

Suppose $a^{17} = e$ and $a^{25} = e$

By prop 1 (i), it follows that $|a| \mid 17$ and $|a| \mid 25$

$\Rightarrow |a| \mid \gcd(17, 25)$ blc "$\gcd(c, d) = sc + td$, some $s, t \in \mathbb{Z}$

i.e. $|a| \mid 1$

$\Rightarrow |a| = 1 \Rightarrow a = e$ blc e is unique (for quiz 一道TF高考题)

Prop 2 Let G be a gp

pg 44 let $a \in G$ s.t $|a| = n$

Some $n \in \mathbb{Z}^+$

let $k \in \mathbb{Z}^+$ Then (i) $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$

(ii) $|a^k| = \frac{n}{\gcd(n,k)}$

Pf: Suppose set-up of (i)

Recall set: A, B, $A = B$ iff $A \subseteq B$, $B \subseteq A$

Call $d = \gcd(n,k)$

"$\subseteq$" let $a^k \in \langle a^k \rangle$

We know $d \mid k$ by def of d

(e) $k = sd$ for some $s \in \mathbb{Z}$

$\therefore a^k = a^{sd}$

$= (a^d)^s$ by assoc $\in \langle a^d \rangle$ by def of $\langle a^d \rangle$

"$\supseteq$" let $a^d \in \langle a^d \rangle$

So $a^d = a^{sn+tk}$, b/c $\gcd(n,k) = sn + tk$ some $s, t \in \mathbb{Z}$

Week 5 June 3

补笔记

Cor: If G is a finite cyclic group. Then $\forall g \in G$, $|g| \mid |G|$

pg65

pf: Suppose G is a cyclic group and $|G| < \infty$

(cyclic group: $\exists a \in G$ s.t. $G = \langle a \rangle$)

Say $|G| = |\langle a \rangle| = n$, for some $n \in \mathbb{Z}^+$

Let $g \in G$. (G is cyclic group, $G = \langle a \rangle$, $g \in G \Rightarrow g = a^k$)

Since $g = a^k$ for some $k \in \mathbb{Z}$ ① b/c $G = \langle a \rangle$

$\therefore |g| = |a^k|$

Prop 2 (ii) $|a^k| = \frac{n}{\gcd(n,k)}$ ⇐ could we use this?

? is the k positive in $|g| = |a^k|$?

Answer: True, we can use Prop 2 (ii), why?

b/c ①: $k \in \mathbb{Z}^+$ ∵ $|G| < \infty$ (G is finite)

$\therefore |g| = |a^k| = \frac{n}{\gcd(n,k)}$

$\Leftrightarrow n = \gcd(n,k)\,|g|$

$\uparrow |G|$ (n is the order of G)

$\gcd(n,k) \in \mathbb{Z}$ by def of $\gcd(n,k)$ b/c $n, k \in \mathbb{Z}^+$

$\therefore |g| \mid |G|$

Cor: Let $n \in \mathbb{Z}^+$. Then $\mathbb{Z}_n = \langle k \rangle$ for some $k \in \mathbb{Z}_n$ $\Leftrightarrow \gcd(n,k) = 1$

pg66

Thm: Fundamental Thm of Cyclic Groups (FTOCG)

pg66

1. Every subgp of a cyclic group is cyclic.
2. Let G be a gp. Let $a \in G$ s.t. $|\langle a \rangle| = n$, some $n \in \mathbb{Z}^+$

   $\forall H \subseteq \langle a \rangle$, $|H| \mid |\langle a \rangle|$
3. Let G be a gp. Let $a \in G$ s.t. $|\langle a \rangle| = n$, some $n \in \mathbb{Z}^+$

   For each positive divisor k of n, we have only one subgp of order k.

   Namely $\langle a^{\frac{n}{k}} \rangle$

Pf: (i) WTS $\forall$ gps G (G is cyclic and $\forall H \leq G \Rightarrow H$ is cyclic)

Suppose G is an arb cyclic gp.

Say $\exists\, a \in G$ s.t. $G = \langle a \rangle$

Let $H \leq G$ WTS H is cyclic (proceed by def)

It follows every element in H is of the form a^k some $k \in \mathbb{Z}$

Week 6 June 8

Recall Prop 2(ii) $|a^k| = \frac{n}{\gcd(n,k)}$

Q1: (a) Find the order of $18 \in \mathbb{Z}_{30}$

(b) $|a| = 24$ find $|a^9|$, $|a^4|$

(c) List the generations of a cyclic gp of order 12.

(a) $\mathbb{Z}_{30} = \langle 1 \rangle$ $|\mathbb{Z}_{30}| = 30$ rewrite $18 = 18 \cdot 1$

$\Rightarrow |18| = \frac{30}{\gcd(30,18)} = \frac{30}{6} = 5$

(b) $|a^9| = \frac{24}{\gcd(24,9)} = \frac{24}{3} = 8$

$|a^4| = \frac{24}{\gcd(24,4)} = 6$

(c) $G = \langle a \rangle$ and $|G| = |a| = 12$. Any element in G is of the form a^k

$|a^k| = \frac{12}{\gcd(12,k)} = 12$ so a^k is a generator $\Leftrightarrow \gcd(12,k) = 1$

$k = \{1, 5, 7, 11\}$

Q2: Given $|a^4| = 6$, find all probabilities for $|a|$

Let $n = |a|$ Then $\frac{n}{\gcd(n,4)} = 6 \Rightarrow n = 6\gcd(n,4)$

Since $\gcd(n,4) \in \{1, 2, 4\}$

① $\gcd(n,4) = 1$ $\Rightarrow n = 6 \cdot 1 = 6$ check $\gcd(6,4) = 2 \neq 1$

② $\gcd(n,4) = 2$ $\Rightarrow n = 6 \cdot 2 = 12$ check $\gcd(12,4) = 4 \neq 2$

③ $\gcd(n,4) = 4$ $\Rightarrow n = 6 \cdot 4 = 24$ check $\gcd(24,4) = 4 = 4$ ✓

Q3: Let G be a gp and $a, b \in G$, Suppose $|a| = m$ and $|b| = n$, with $\gcd(m,n) = 1$

(a) prove that $\langle a \rangle \cap \langle b \rangle = \{e\}$

(b) Suppose $ab = ba$ Prove $|ab| = mn$

(a)

(b) $(ab)^{mn} = a^{mn}b^{mn} = (a^m)^n (b^n)^m = e^n e^m = e$
(commutability; assoc; commutability)

$\Rightarrow |ab| \mid mn$ by Prop 1 (i)

Let $|ab| = d$ Then $(ab)^d = e \Rightarrow a^d b^d = e \Rightarrow a^d = b^{-d}$

$\Rightarrow a^d \in \langle a \rangle$ and $a^d \in \langle b \rangle \Rightarrow a^d \in \langle a \rangle \cap \langle b \rangle = \{e\}$, from (a)

$\therefore a^d = e$ and $b^d = e \Rightarrow m \mid d$ and $n \mid d$ Since $\gcd(m,n) = 1$

$\Rightarrow mn \mid d = |ab|$ Therefore, $|ab| = mn$.

Q4 $\alpha = (1,2,3)(2,4,1)(3,7,8,6) \in S_8$ Compute α^{103}

Notes that disjoint cycles commute.

first write as disjoint cycles

$\alpha = (1,3,7,8,6)(2,4)$ $\quad |(1,3,7,8,6)| = 5$

$|(2,4)| = 2$

$\alpha^{103} = [(1,3,7,8,6)(2,4)]^{103}$

$= (1,3,7,8,6)^{103}(2,4)^{103}$ as disjoint cycles commute.

$= (1,3,7,8,6)^{5\cdot 20+3}(2,4)^{2\cdot 51+1}$

$= ((1,3,7,8,6)^5)^{20}(1,3,7,8,6)^3 ((2,4)^2)^{51}(2,4)$

$= (1,3,7,8,6)^3 (2,4)$

$= (1,8,3,6,7)(2,4)$

Lec

Recall FTOCG

1. Every subgp of a cyclic gp is cyclic

2. Let $n \in \mathbb{Z}^+$ Suppose $|\langle a \rangle| = n$, if H is any subgp of $\langle a \rangle$ then $|H| \mid n$

3. Let $n \in \mathbb{Z}^+$ Suppose $|\langle a \rangle| = n$

For each positive divisor k of n there is a unique subgp of $\langle a \rangle$ of order k, ① Moreover these are all the subgps of $\langle a \rangle$

① Namely, $\langle a^{\frac{n}{k}} \rangle$, two things to prove: ② the order of $k = |\langle a^{n/k} \rangle|$

Pf of 2: Let $n \in \mathbb{Z}^+$ and let $|\langle a \rangle| = n$.

Suppose $H \leq \langle a \rangle$ is arb.

· If $H = \{e\}$, then done ✓

· If $H \neq \{e\}$, we know from pf of 1., $H = \langle a^m \rangle$

where m is the smallest positive integer s.t. $a^m \in H$

So $|H| = |\langle a^m \rangle| = |a^m| = \frac{n}{\gcd(m,n)}$

(↑ prop 1 (ii)) (↑ prop 2 (ii))

It follows $n = \gcd(m,n)\,|H|$

($\langle a \rangle$ = ; $\gcd(m,n) \in \mathbb{Z}$ b/c $m, n \in \mathbb{Z}^+$)

$\therefore |H| \mid n$ ☑

Pf of 3: Let $n \in \mathbb{Z}^+$ and let $|\langle a \rangle| = n$

Let k be any positive divisor of n

① $|\langle a^{\frac{n}{k}} \rangle| = k$

Consider $|\langle a^{\frac{n}{k}} \rangle| = |a^{\frac{n}{k}}|$ by "$|\langle a \rangle| = |a|$" Prop 1 (ii).

$$= \frac{n}{\gcd(n/k, n)} \quad \text{by Prop 2 (ii)}$$

$$= \frac{n}{\frac{n}{k}}$$

$$= k$$ ☐

② Show $\langle a^{\frac{n}{k}} \rangle$ is the only subgp of order k

By set up $|\langle a \rangle| = n$, by def of order of element $a^n = e$

if $k = 1 \Rightarrow \langle a^{\frac{n}{1}} \rangle = \langle a^n \rangle = \langle e \rangle$

If $k \neq 1$, so $\langle a^{\frac{n}{k}} \rangle \neq \{e\}$

Suppose $H \leq \langle a \rangle$ w. order k and <u>not trivial</u>

b/c we know $H = \langle a^m \rangle$ where m is the smallest trivial integer s.t $a^m \in H$

Consider $\langle a^m \rangle = \langle a^{\gcd(n,m)} \rangle$ by prop 2 (i)

but in pf of 2. we saw $\gcd(m,n) = \frac{n}{|H|} = \frac{n}{k}$ (b/c $|H| = k$)

$= \langle a^{\frac{n}{k}} \rangle$ ☐

Cor: For each positive divisor k of n, $\mathbb{Z}_n$ has a unique subgp of order k. Namely $\langle \frac{n}{k} \rangle$. Moreover these are all the subgp of $\mathbb{Z}_n$.

pf: Take $G = \mathbb{Z}_n$ and use FTOCG § (+2) ☐

Week 6 June 10

Ex: List all the subgps of $\mathbb{Z}_{21}$

Sol: Recall $\mathbb{Z}_{21} = \langle 1 \rangle$ so he is cyclic

As $|\mathbb{Z}_{21}| = 21 = 3 \cdot 7$

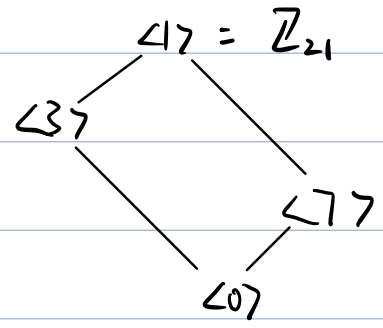
We know the positive divisors of 21 are 1, 3, 7, 21

By FTOCG (Cor) we have the subgps

- $\langle \frac{21}{21} \rangle = \langle 1 \rangle = \mathbb{Z}_{21}$, this is a subgp of order 21
- $\langle \frac{21}{7} \rangle = \langle 3 \rangle = \{3, 6, 9, 12, 15, 18, 0\}$ ✓ of order 7
- $\langle \frac{21}{3} \rangle = \langle 7 \rangle = \{0, 7, 14\}$ ✓ of order 3
- $\langle \frac{21}{1} \rangle = \langle 0 \rangle = \{0\}$ trivial subgp / subgp of order 1

Def: Lattice diagram or subgp diagram of a gp G, is a pictoral representation of G, it's subgp and their relationships.

eg lattice diagram of $\mathbb{Z}_{21}$:



3 different height

trivial always bottom.

trivial subgp always subgp of ⟨3⟩ ⟨7⟩

End for Quiz 2.

2R => 18 17
∞ 15
∞ 15

20 red 16 blue 20 16 2 1

same color add blue 36 . . o o . o

Ch 5, Permutations

Recall we have met the symmetric gp S_n of degree n, these are the permutations of $X = \{1, 2, \dots n\}$ for some $n \in \mathbb{Z}^+$ w. composition

- Cycle Notation

e.g. $\alpha = (1 \ 2 \ 3) \in S_3$

$\alpha(1) = 2$

$\alpha(2) = 3$

$\alpha(3) = 1$

- def of a k-cycle or the length of a cycle.
- $|S_n| = n!$

Def we say cycles $\alpha, \beta \in S_n$ for some $n \in \mathbb{Z}^+$ a <u>disjoint</u> if $a_i \neq b_j \ \forall i, j$ where a_i's are elements in α and b_j's are elements in β.

- α and β have no common elements
- $\alpha = (1, 2, 4)$ in S_6

  $\beta = (3, 5)$

  these are disjoint.

Ex: Let $\alpha \in S_6$ s.t. $\alpha = (2, 4, 5)(6, 1, 2)(5, 3, 2, 4)(4, 5)$

express this as a product of disjoint cycles

Soln: $\alpha = (1, 4, 3, 6)(2, 5)$ // by expanding.

$= (2, 5)(1, 4, 3, 6)$

Thm. Let $n \in \mathbb{Z}^+$ Every permutation in S_n is either a k-cycle or can be expressed as a product of disjoint cycles.

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#4

Thm: Disjoint cycles commute.

i.e. Let $n \in \mathbb{Z}^+$ Let $\alpha, \beta \in S_n$

If α, β disjoint then $\alpha\beta = \beta\alpha$

We can write $X = \{1, 2, \cdots n\}$ by partitioning him as follows

$\{\underbrace{a_1, a_2, a_3 \cdots a_m}_{\text{fixed only by } \beta}, \underbrace{b_1, b_2 \cdots b_\ell}_{\text{fixed only by } \alpha}, \underbrace{c_1 \cdots c_n}_{\text{fixed only by } \alpha \text{ and } \beta}\}$

WTS $(\alpha \circ \beta)(x) = (\beta \circ \alpha)(x, \forall x \in X)$

- Let $a_i \in X$ be arb.

Consider $(\alpha \circ \beta)(a_i) = \alpha(\beta(a_j))$

$= \alpha(a_i)$ blc β fixes the a_i's

$= a_{i+1}$ by def α

and $(\beta \circ \alpha)a_i = \beta(\alpha(a_i))$

$= \beta(a_{i+1})$ by def of α

$= a_{i+1}$ blc β fixes the a_i's

- Let $b_j \in X$ be arb.

So $\alpha(\beta(b_j)) = \alpha(b_{j+1})$ by def β.

$= b_{j+1}$ blc α fixes the b_j's

and $\beta(\alpha(b_j)) = \beta(b_j)$ - - -

$= b_{j+1}$ - - -

- Let $c_k \in X$ be arb

So $\alpha(\beta(c_k)) = \alpha(c_k)$ blc β fix c_k

$= c_k$ blc α fix c_k

and $\beta[\alpha(C_k 1)] = \beta(C_k)$ - - -

$\geq C_k$ - - - -

Thm: "LCM-order Thm"

Let take $n \in \mathbb{Z}^+$, Let $\alpha \in S_n$

If α is a k-cycle or is expressed as a product of disjoint cycles.

Say $\alpha = \alpha_1 \alpha_2 \cdots \alpha_n$.

then $|\alpha| = lcm\{l_1, l_2 \cdots l_m\}$, where each l_i is the length of α_i

(e.g.) $\alpha = (1,2)(3,4,5)$, what is $|\alpha| = ?$

This decomp is disjoint

∴ by lcm-thm, $|\alpha| = lm\{2,3\} = 6$ //

Pf of "lcm" thm

Let $n \in \mathbb{Z}^+$ Let $\alpha \in S_n$

Suppose α is a k-cycle or $\alpha = \alpha_1 \alpha_2 \cdots \alpha_n$

Where α_i's are (pairwise) disjoint

· l_i is the length of α_i for each i.

1) if α is a k-cycle, say $\alpha = (a_1 \cdots a_k)$ then $|\alpha| = l_k$

2) if $\alpha = \alpha_1 \cdots \alpha_m$

Note $\exists j \in \mathbb{Z}^+$ s.t $\alpha^j = (1)$ b/c $|S| < \infty$

for some $j \in \mathbb{Z}^+$

Consider $\alpha^j = (1) \iff \underbrace{(\alpha_1 \alpha_2 \cdots \alpha_m)(\alpha_1 \alpha_2 \cdots \alpha_m) \cdots (\alpha_1 \alpha_2 \cdots \alpha_m) = (1)}_{j \text{ term}}$

commute

Week 7 TUT

1. Let $\sigma = (1,5,2,6,3)(2,4,7,5) \in S_7$, σ is disjoint form, give cycle type and order, then compute σ^{59}

Soln: Rewrite $\sigma = (1,5,6,3)(2,4,7)$ Cycle types $(4,3)$

$|\sigma| = \text{lcm}(4,3) = 12$. To compute σ^{59}, apply dimension algorithm, $59 = 4 \times 12 + 11$

Thus, $\sigma^{59} = \sigma^{11} = \sigma^{-1} = ((1,5,6,3)(2,4,7))^{-1}$

$= (2,4,7)^{-1}(1,5,6,3)^{-1}$

$= (7,4,2)(3,6,5,1)$

$= (3,6,5,1)(7,4,2)$ b/c disjoint cycle commute.

Q2 (a) How many elements of S_6 have order 6?

(b) How many of these lie in A_6

(a) Find valid order types: What integer partitions have lcm of 6.

① 6-cycles

② 3-cycle with 2 cycle (length 3,2,1)

All others fails. e.g. $3+3=6$ $\text{Lcm}(3,3)=3$

① 6-cycle: $\frac{k!}{k} = (k-1)! = 120$

② Pick 3 numbers from 6 elements $\binom{6}{3} = 20$

From 3 cycle $(3-1)! = 2$

Pick 2 element from remaining 3 $\binom{3}{2} = 3$

From 2 cycle $(2-1)! = 1$

$20 \times 2 \times 3 \times 1 = 120$

Total $120 + 120 = 240$.

(b) 6-cycle decomposes into 5 transpositions $\Rightarrow$ odd

Cycle type $(3,2,1)$ has $(3-1)+(2-1) = 3$ transpositions $\Rightarrow$ odd

Neither of them are in A_6

0 element has order 6 in A_6

Q3: Let $\alpha \in S_n$ where $n \geq 2$ Prove that if α is of odd order, then $\alpha \in A_n$

Pf: Let $|\alpha| = 2k+1$ for some $k \in \mathbb{Z}^+$ This means $\alpha^{2k+1} = (1) = \underbrace{(1\ 2)(2\ 1)}_{\text{even}}$

Suppose α can be decomposed into m transpositions. Then α^{2k+1} can be decomposed into $m(2k+1)$ transposition, which must be even from the previous equation. But $2k+1$ is odd So m must be even. Therefore, $\alpha \in A_n$

□

Q4. Let $\beta = (b_1, b_2, \cdots b_r) \in S_n$ be a r-cycle. Prove that for any $\alpha \in S_n$. $\alpha\beta\alpha^{-1}$ is the r-cycle $(\alpha(b_1), \alpha(b_2) \cdots \alpha(b_r))$

Pf: Let $a_i = \alpha(b_i)$ for $i = 1, \cdots r$ WTS $\alpha\beta\alpha^{-1}$ maps $a_1 \to a_2 \to \cdots \to a_r \to a_1$ and it fixes everything else.

① elements inside cycle.

$\alpha\beta\alpha^{-1}(a_i) = \alpha(\beta(\alpha^{-1}(a_i))) = \alpha(\beta(b_i))$ if $i < r$ then $\beta(b_i) = b_{i+1}$, $\alpha(b_{i+1}) = a_{i+1}$
if $i = r$ then $\beta(b_i) = b_i$, $\alpha(b_i) = a_i$

② elements outside cycle

Let c be an elements where $c \neq a_i$ for all i This means $c \neq \alpha(b_i)$ for all

As α is a bijection, $b_i \neq \alpha^{-1}(c)$ Therefore, $\beta(\alpha^{-1}(c)) = \alpha^{-1}(c)$

So $\alpha\beta\alpha^{-1}(c) = \alpha(\alpha^{-1}(c)) = c$

Exercise: ① Prove or disprove: every permutation of even order is an odd permutation.

② Let $\alpha = (1, 5, 3, 6)$ and $\beta = (2, 4)(5, 7) \in S_7$ Is $\alpha^2\beta$ in A_7?

What is its order?

Lec Ch6 + Ch 10

Homomorphisms & Isomorphisms

Motivation, consider $H = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in \mathbb{R} \right\}$ this is a gp under mult

consider $\begin{pmatrix} 1 & b_1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & b_2 \\ 0 & 1 \end{pmatrix} \in H$ so $b_1, b_2 \in \mathbb{R}$

$$\begin{pmatrix} 1 & b_1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & b_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b_1+b_2 \\ 0 & 1 \end{pmatrix}$$

we know $(\mathbb{R}, +)$ is a gp

Define $\phi : (\mathbb{R}, +) \to (H, \cdot)$

$$b \to \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

We can check:

$$\phi(b_1+b_2) = \phi(b_1) \cdot \phi(b_2) \quad \text{any } b_1, b_2 \in \mathbb{R}$$

($+$: bin op in $\mathbb{R}$; $\cdot$: bin op in H)

This map is an example of a homo

Def: Let $G, \bar{G}$ be gps, A map $\phi : G \to \bar{G}$ is a homomorphism if

$$\forall a, b \in G, \quad \phi(a \cdot b) = \phi(a) \cdot \phi(b)$$

($\cdot$: bin op in G; $\cdot$: bin op in $\bar{G}$)

Iso Def: A map $\phi : G \to \bar{G}$ is an isomorphisms if ϕ is a homo and ϕ is bijection. When an iso $\phi : G \to \bar{G}$ exists we say "G and $\bar{G}$ are isomorphic" Denoted $G \cong \bar{G}$

Example: Let $G, \bar{G}$ are gps.

. The map $\phi : G \to \bar{G}$ defined by $\phi(g) = \bar{e}$, $\forall g \in G$ is a hom. It's called the trivial homomorphisms.

Pf: let $g_1, g_2 \in G$. Consider $\phi(g_1, g_2) = \bar{e}$ by def of ϕ

$g_1, g_2 \in G$ (b/c G is a gp (Closure))

$= \bar{e}\bar{e} = \phi(g_1)\phi(g_2)$ by def of ϕ

∎

• The $\phi : G \to G$ defined by $\phi(g) = g$, $\forall g \in G$ is a homomorphism. It's called the identity homomorphism. (identity map is always bijection)

In fact, this map is an isomorphism. This is called the trivial isomorphism.

• If $H \leq G$, the map $\phi : H \to G$ defined by $\phi(h) = h$, $\forall h \in H$.

This is a hom, It's called the inclusive homo

Ex: Consider $\phi : GL(2,\mathbb{R}) \to GL(2,\mathbb{R})$ $A \to A^2$

is ϕ a homo?

pairwise disjoint cycles commute.

Def Let $n \in \mathbb{Z}^+$ and let $\alpha \in S_n$ s.t. $\alpha = \alpha_1 \cdots \alpha_m$ is a disjoint decomposition of α where l_i is the length of α_i. The <u>cycle type of α</u> is a list in decreasing order of the l_i's.

Ex Consider S_4. $X = \{1,2,3,4\}$

- What are the possible cycle types in S_4?
- List the elements in each cycle type.

SOLN:

cycle types	elements in S_4 of that cycle type
$(1,1,1,1)$	$(1)(2)(3)(4) = (1)$
$(2,1,1)$	$(1\,2)$ $(1\,3)$ $(1\,4)$ $(2\,3)$ $(2\,4)$ $(3\,4)$
$(2,2)$	$(1,2)(3,4)$ $(1,3)(2,4)$, $(1,4)(2,3)$
$(3,1)$	$(1,2,3)$ $(1,2,4)$ $(1,3,4)$ $(2,3,4)$
(4)	$(1,2,3,4) \times 6$

$(1,3,2,4)$ $(1,4,2,3)$ $(1,4,3,2)$ all in !

Ex Find an element of order 10 in S_7.

SOLN: we seek $\alpha \in S_7$ s.t. $|\alpha| = 10 = 5 \cdot 2$

(i@) want $\alpha \in S_7$ with cycle type $(5,2)$

b/c $\operatorname{lcm}\{5,2\} = 10 = |\alpha|$

$$\alpha = (1\,2\,3\,4\,5)(6\,7)$$

Exercise: How many elements in S_7 have order 10?

Hint: seek all $\alpha \in S_7$ w. cycle type (5,2)

How many $\alpha \in S_7$ w. $\alpha = (\quad)(\quad)$
5-cycle 2-cycle
↑ ↗
disjoint!

Writing proofs ✡✡ e.g. disjoint cycle commute.

Thm: Every permutation is S_n $(n>1)$ can be written as a product of 2-cycles (transpositions).

Pf: Let $n \in \mathbb{Z}^+$, $n>1$. Consider S_n. $(X = \{1,2,\dots,n\})$

- Note $(1) = (12)(21)$ ✓
- Let $\alpha \in S_n$ be arb. st. $\alpha \neq (1)$.

We know α can written as

$$\alpha = \alpha_1 \alpha_2 \alpha_3 \cdots \alpha_m \quad (*)$$

where α_i's are disjoint.

It's sufficient to show α_i can be written as a product of transpositions

(any α_i in $(*)$, $1 \le i \le m$)

Note α_i is an i-cycle

so $\alpha_i = (a_1\ a_2 \cdots a_i)$ where $a_j \in X$



we can directly compute

$$(a_1\ a_i)(a_1\ a_{i-1}) \cdots (a_1\ a_3)(a_1\ a_2)$$
$$= (a_1\ a_2\ a_3 \cdots a_i)$$
as needed

Fast eg.

(#9) $\alpha = (1234) \in S_5$

$\alpha = (14)(13)(12)$

$= (234)$

opp

∎

Parity of S_n $(n>1)$

Def

- A permutation is **even** if it can be written as an even number of transpositions
- A permutation is **odd** " " " " "

" " odd " " "

Ex What is the parity of each cycle type in S_4?

SOLN:

cycle types	parity
(4)	odd
$(3,1)$	even
$(2,2)$	even
$(2,1,1)$	odd
$(1,1,1,1)$	even

eg $(1\,2\,3\,4) = (1\,4)(1\,3)(1\,2)$

eg $(1\,2\,3) = (1\,3)(1\,2)$

eg $(1) = (1\,2)(2\,1)$

Thm: The set $(n>1)$

$$A_n = \{\alpha \in S_n \mid \alpha \text{ is even}\}$$

is a subgp of S_n.

Pf: exercise. ∎

Def A_n is called the **alternating gp of degree n**

Thm: Let $n \in \mathbb{Z}^+$, $n > 1$.

Then $|A_n| = \frac{n!}{2}$

Pf: Consider S_n, $n > 1$

Consider

$$H: \{\text{even perm. in } S_n\} \longrightarrow \{\text{odd perm. in } S_n\}$$

$$A_n \qquad \alpha \longmapsto \alpha(12) \quad \text{(ie) } H(\alpha) = \alpha(12)$$

Show H is a bijection!

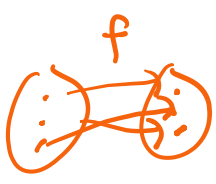
- injective: Let $\alpha, \beta \in A_n$

Suppose $H(\alpha) = H(\beta)$

$\Rightarrow \alpha(12) = \beta(12)$ by H def.

$\Rightarrow \alpha = \beta$ by right-cancellation law

$\therefore H$ is 1-1 ✓



$\forall a, b \in \text{dom}(f)$
$f(a) = f(b)$
$\Rightarrow a = b$

- onto: Let $\beta \in \{\text{odd perm in } S_n\}$.

Choose $\alpha = \beta(12) \in A_n$

$f: A \to B$
$\forall b \in B, \exists a \in A$
s.t. $f(a) = b$.

Then $H(\alpha) = \alpha(12)$ by H formula

$= \beta(12)(12)$ by α choice

$= \beta(1)$

$= \beta$ as wanted

$\therefore H$ is a bijection

Note $S_n = \{\text{even perm}\} \sqcup \{\text{odd perm}\}$

$\uparrow$ disjoint

$\Rightarrow |S_n| = |\{\text{even perm}\}| + |\{\text{odd perm}\}|$

$n!$

but $|\{\text{even perm.}\}| = |\{\text{odd perm.}\}|$ b/c H bijective

$= 2|A_n|$

(ie) $n! = 2|A_n|$

$\therefore |A_n| = \frac{n!}{2}$ as needed

//

(3)

June 24.

last day,

Ex. Is $\phi: GL(2,\mathbb{R}) \to GL(2,\mathbb{R})$, $A \mapsto A^2$ a hom.?

SOLN: No, Consider $A=\begin{pmatrix}1&1\\0&1\end{pmatrix}, B=\begin{pmatrix}1&0\\1&1\end{pmatrix}$. Note $A,B \in GL(2,\mathbb{R})$ b/c $\det(A)=1\neq 0$, $\det(B)=1\neq 0$

We compute $AB=\begin{pmatrix}2&1\\1&1\end{pmatrix}$

Then $\phi(AB)=\begin{pmatrix}2&1\\1&1\end{pmatrix}^2=\begin{pmatrix}5&3\\3&2\end{pmatrix}$

But $\phi(A)\phi(B)=\begin{pmatrix}1&1\\0&1\end{pmatrix}^2\begin{pmatrix}1&0\\1&1\end{pmatrix}^2$

$=\begin{pmatrix}3&2\\2&1\end{pmatrix}\begin{pmatrix}3&2\\2&1\end{pmatrix}=\begin{pmatrix}13&8\\8&5\end{pmatrix}$

$\neq$

$\therefore \phi(AB)\neq\phi(A)\phi(B)$ ☒

Thm: Let $G, \bar{G}$ be gps Let $\phi: G \to \bar{G}$ be a hom

The set

$$\ker(\phi)=\{g\in G \mid \phi(g)=\bar{e}\}\subseteq G$$

is a subgp of G

This is the kernel of ϕ

Pf: We apply the 1-step subgp test

Note. • $\ker\phi \subseteq G$ by def of $\ker(\phi)$

• Consider $\phi(e)$, remember $e\in G$ b/c G is gp

So $\phi(e)=\phi(ee)=\phi(e)\phi(e)$

↳ b/c ϕ is homo

Equivalently $\bar{e}\,\phi(e)=\phi(e)\phi(e)$

$\Rightarrow \bar{e}=\phi(e)$ by right cancellation law

$\therefore e\in\ker(\phi)$ Hence $\ker\phi\neq\emptyset$

Let $g_1, g_2 \in \ker\phi$ So $g_1, g_2 \in G$ and $\phi(g_1)=\bar{e}=\phi(g_2)$ by $\ker\phi$ def.

Consider $g_1 g_2^{-1}$:

• $g_2\in G \Rightarrow g_2^{-1}\in G$ by inverse axiom of G

$\Rightarrow g_1 g_2^{-1}\in G$, by closure of G.

• $\phi(g_1 g_2^{-1})=\phi(g_1)\phi(g_2^{-1})$ b/c ϕ is homo.

$=\phi(g_2)\phi(g_2^{-1})$ by $\phi(g_1)=\bar{e}=\phi(g_2)$

$=\phi(g_2 g_2^{-1})$ b/c ϕ is a homo

$=\phi(e)$

$=\bar{e}$ by above prior non-empty work.

$\therefore g_1 g_2^{-1}\in\ker(\phi)$

$\therefore \ker\phi\leq G$

☒

Thm: Let $G, \bar{G}$ be gps, Let $\phi: G \to \bar{G}$ be a homo

Then (i) $\phi(e) = \bar{e}$

(ii) $\forall n \in \mathbb{Z}, \forall g \in G \quad \phi(g^n) = (\phi(g))^n$

(iii) $\forall g \in G, (|g| < \infty \Rightarrow |\phi(g)| \mid |g|)$

(iv) ϕ is injective $\Leftrightarrow \ker(\phi) = \{e\}$

Pf: Suppose set-up

(i) we did it last day

(ii) Let $n \in \mathbb{Z}$, $g \in G$

- $n > 0$ so $\phi(g^n) = \phi(g \cdot g \cdots)$

$= \underbrace{\phi(g)\,\phi(g) \cdots \phi(g)}_{n \text{ times}}$, by def of ϕ (n times)

$= (\phi(g))^n$ (a)

$\therefore \phi(g^0) = \phi(e) = \bar{e} = (\phi(g))^0$

- $n < 0 \quad \phi(g^n) = \phi(g^{-1} \cdot g^{-1} \cdots g^{-1})$

$= (\phi(g^{-1}))^{|n|}$, by def of hom ($|n|$ times)

Since we know $\phi(e) = \bar{e}$

$\Leftrightarrow \phi(g)\phi(g^{-1}) = \bar{e}$ blc ϕ is a homo

$\Rightarrow \phi(g^{-1}) = (\phi(g))^{-1}$ blc G is a gp (inverse axiom, assoc in $\bar{G}$ of id axiom)

$\to (\phi(g^{-1}))^{|n|} = (\phi(g))^n$

Pf: (iii) Let $g \in G$

Suppose $|g| = n < \infty$, some $n \in \mathbb{Z}^{>0}$

WTS $|\phi(g)| \mid n$

Consider $(\phi(g))^n = \phi(g^n)$ by (ii)

$= \phi(e)$ blc $|g| = n$

$= \bar{e}$ by (i)

$\Rightarrow |\phi(g)| \mid n$, as needed

(iv) WTS ϕ is $1-1 \iff \ker\phi = \{e\}$

$\Rightarrow$ Suppose ϕ is injective

WTS $\ker(\phi) = \{e\}$, Let $g_1, g_2 \in \ker\phi$.

So $g_1, g_2 \in G$ and $\phi(g_1) = \bar{e} = \phi(g_2)$ by $\ker\phi$ def.

$\Rightarrow \phi(g_1)(\phi(g_2))^{-1} = \phi(g_2)(\phi(g_2))^{-1}$

$\phi(g_2^{-1})$ $\qquad$ $\phi(g_2^{-1})$ by (ii)

为什么这里要这样设.

$\iff \phi(g_1 g_2^{-1}) = \phi(g_2 g_2^{-1})$ by ϕ homo def

$= \phi(e)$

$= \bar{e}$, by homo def. (i)

$\therefore g_1 g_2^{-1} \in \ker\phi$, by def of $\ker\phi$ and $g_1 g_2^{-1} \in G$

($g_1, g_2 \in G$ by def of $\ker\phi$)

$g_1 g_2^{-1} = e$ b/c $\ker(\phi) = \{e\}$

$g_1 = g_2$ $\therefore \phi$ is injective.

$\Leftarrow$ Suppose $\ker\phi = \{e\}$ wts $1-1$

Let $g_1, g_2 \in G$

Suppose $\phi(g_1) = \phi(g_2)$

$\Rightarrow$ Suppose ϕ is $1-1$ wts $\ker\phi = \{e\}$

We know $\phi(e) = \bar{e}$ by (i)

But ϕ is $1-1$ $\therefore e$ is the only element that can map to $\bar{e}$

i.e. $\ker\phi = \{e\}$ as needed.

Thm: Let $G, \bar{G}$ be gps. Let $\phi: G \to \bar{G}$. The set $\phi(G) = \{\phi(g) \in \bar{G} \mid g \in G\}$ this is the image of ϕ.

Then (i) $\phi(G) \leq \bar{G}$

(ii) If G is cyclic $\iff$ then $\phi(G)$ is cyclic

(iii) If G is abelian then $\phi(G)$ is abelian.